

TraceFund

Blockchain-Based Transparent Fund Disbursement and Fraud Detection System

Group No: 19

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Abstract

Public funds such as government subsidies, disaster relief, and welfare payments are currently disbursed as cash or standard bank transfers. Once this money reaches a beneficiary, the government has no practical way to control, verify, or trace how it is ultimately spent, and no reliable audit trail exists to detect diversion or misuse. This lack of visibility and enforcement is a major contributor to fund leakage and corruption in welfare and relief programs.

Trace-Fund addresses this problem at its root by changing the nature of the money itself rather than monitoring it after the fact. Public funds are issued as programmable digital tokens with spending rules built directly into them, enforced automatically through smart contracts on a blockchain network. When a beneficiary attempts to spend a token, the smart contract instantly checks whether the receiving party is a verified vendor within the approved category, such as food or medicine, and approves or rejects the transaction within seconds, with no manual intervention or opportunity for discretionary interference. Every transaction is recorded permanently on the blockchain, creating an immutable, tamper-proof audit trail from the moment funds are issued to the moment they are redeemed.

A graph-based fraud detection layer continuously analyzes transaction flows across wallets to identify structural patterns associated with fraud, such as layering funds across intermediary accounts or mule-account behavior, flagging suspicious activity in near real time rather than after the fact. A public transparency dashboard further displays aggregate, category-wise fund utilization, such as total disbursement and the percentage spent on food versus medicine, allowing independent verification by citizens and civil society without ever exposing any individual beneficiary's identity or personal transaction history.

By combining programmable money, automated rule enforcement, immutable record-keeping, and proactive fraud detection into a single cohesive system, Trace-Fund offers a practical, technically grounded approach to reducing corruption in public fund disbursement, conceptually aligned with real-world initiatives such as the Reserve Bank of India's e-Rupee (CBDC) pilot.

Technologies Used

Frontend

- **React.js**
Good if you want a simpler single-page app for the wallet/vendor/dashboard interfaces without needing server-side rendering or routing complexity. Lighter to set up for a project-scope build.
- **Next.js**
What you already have listed. Good choice if you want built-in routing, API routes, and easy deployment (e.g., on Vercel) — useful since you'll have separate dashboards for Government, Vendor, Beneficiary, and Public.

Backend

- **Python:** Core backend programming language
- **FastAPI:** Backend web framework used to build REST APIs
- **Web3.py:** Python library used to connect the backend to the blockchain network and smart contracts
- **Solidity:** Language used to write smart contracts that enforce spending restrictions
- **Polygon / Ethereum Testnet:** Blockchain network on which smart contracts are deployed
- **PostgreSQL:** Off-chain database used to store vendor details, categories, and other non-sensitive data